

Environment

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Environment ka concept, components/types, degradation ke causes, aur India mein protection ka legal/institutional framework — is chapter ka core.
- Organisations, Movements, aur Important Days — ye teeno high-yield one-liner zones hain.

✓ QUICK RECALL

"Environment" = Environ (surrounding) + ment → ek covering jo humein chaaro taraf se ghere rehti hai.

2. Components & Types — BAE Trick

- Biotic → Living beings, animals, plants, microbes.
- Abiotic → Water, air, soil, etc.
- Energy → Solar, geothermal, etc.

🧠 MNEMONIC — BAE (Biotic-Abiotic-Energy)

Environment ka "BAE" (best friend) teen cheezein hain: Biotic, Abiotic, Energy — teenon alag-alag components hain, Energy inme se kisi mein club nahi hoti.

⚠️ EXAM TRAP

Energy (solar, geothermal) ko galti se Abiotic ke andar daal diya jaata hai. Reality: Energy ek teesra, DISTINCT component hai — Biotic aur Abiotic ke barabar, unke andar nahi.

- Natural Environment → Land, air, water, etc.
- Man-made Environment → Agriculture, industrial cities, etc.

3. Degradation & Protection — 1986/2010 Trick

- Environmental degradation = environmental imbalance; causes: Mining, Industrialization, Urbanization, Modern agriculture.
- Environment (Protection) Act, 1986 → isi se Genetic Engineering Approval Committee bana.
- National Green Tribunal (NGT), 2010 → environment matters ke speedy resolution ke liye.
- Central Pollution Control Board (CPCB) → key regulatory body.

🧠 MNEMONIC — "86 se Act, 10 se Tribunal"

1986 → Act banta hai (kanoon), 2010 → Tribunal banta hai (fast justice). Pehle rule banta hai, phir uske 24 saal baad usko enforce karne wali fast-track court banti hai.

✓ QUICK RECALL

Agenda-21 → Rio Earth Summit (1992) mein adopt hua, aim tha ecological destruction aur economic inequality kam karna.

4. Organisations — WNGU Chain (HQ Trick)

Organisation	Est.	HQ
World Meteorological Org. (WMO)	1950	Geneva (Switzerland)
NEERI	1958	Nagpur (Maharashtra)
Greenpeace International	1971	Amsterdam (Netherlands)

Organisation	Est.	HQ
UNEP	1972	Nairobi (Kenya)

MNEMONIC — WNGU — "50-58-71-72" chronology

W(50) → N(58) → G(71) → U(72): WMO sabse purana (Geneva), phir NEERI (Nagpur), phir Greenpeace (Amsterdam), sabse aakhir mein UNEP (Nairobi). Saal badhte order mein hi HQ yaad karo: Geneva → Nagpur → Amsterdam → Nairobi.

EXAM TRAP

UNEP ka HQ Nairobi (Kenya) hai, NAHI ki Geneva/New York — jo default UN-HQ guess hota hai. UNEP un chuninda major UN bodies mein se ek hai jinka HQ Africa mein hai.

5. Movements — "Chipko ka Chhota Bhai" Trick

Movement	Region	Leader(s)
Chipko Movement	Uttarakhand	Sunderlal Bahuguna & Chandi Prasad Bhatt
Appiko Movement (1983)	Karnataka	Pandurang Hegde
Narmada Bachao Andolan (1985)	Narmada Valley	Medha Patkar
Matti Movement (1996)	Chamoli (Uttarakhand)	Kalyan Singh Rawat

QUICK RECALL

Appiko (Karnataka, 1983) ko "southern version" of Chipko kaha jaata hai — same tree-hugging concept, bas region (Karnataka) aur leader (Pandurang Hegde) alag hain.

EXAM TRAP

Matti Movement (1996) bhi Chamoli, Uttarakhand se hai — Chipko se region same hone ki wajah se dono confuse ho jaate hain. Leader se differentiate karo: Chipko = Bahuguna/Bhatt, Matti = Kalyan Singh Rawat.

6. Important Days — "No Fixed Date" Trick

Day	Date
World Wildlife Day	3rd March
World Forest Day	21st March
World Earth Day	22nd April
World Environment Day	5th June
International Tiger Day	29th July
World Ozone Day	16th September
World Habitat Day	First Monday of October

MNEMONIC — "3-21-22-5-29-16" ladder

March mein 2 din hain (3, 21), phir April (22), June (5), July (29), September (16) — ek hi seedhi mein chadhte jao. Sirf October ka Habitat Day is ladder se BAHAR hai (floating date).

EXAM TRAP

World Habitat Day ki koi FIXED calendar date nahi hai — ye har saal "First Monday of October" ko padta hai, jabki baaki sabhi days ki fixed dates hain (3 March, 21 March, etc.).

Ecomark (1991) → Eco-friendly products ke liye label.

 MNEMONIC — "87 Indira, 99 Rajiv"

Indira Gandhi Environmental Award → 1987. Rajiv Gandhi Environmental Award → 1999. Maa pehle (87), beta baad mein (99) — jaise real life mein bhi Indira pehle, Rajiv baad mein aaye.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Energy environment ka component hai ya Abiotic ka part?	Energy Biotic/Abiotic se ALAG, teesra distinct component hai
1986 Act ya 2010 NGT — kaunsa pehle?	1986 = Environment (Protection) Act; 2010 = NGT (baad mein bana)
UNEP ka HQ Geneva/New York?	UNEP HQ = Nairobi, Kenya (Africa mein)
Matti Movement kaunsi state se hai?	Chamoli, Uttarakhand (Chipko se region same, leader alag — Kalyan Singh Rawat)
World Habitat Day ki fixed date kya hai?	Koi fixed date nahi — "First Monday of October"
Indira ya Rajiv Award pehle aaya?	Indira Gandhi Award = 1987 (pehle); Rajiv Gandhi Award = 1999 (baad mein)
Appiko Movement kis state ka "southern Chipko" hai?	Karnataka (1983), leader Pandurang Hegde

Ecosystem

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Ecosystem = environment aur biotic community ke beech ke natural relationships ka study.
- Cover hota hai: components & types, food chain & trophic levels, ecological pyramids, ecological niche, imbalance, biogeochemical cycles, biomes.

✓ QUICK RECALL

Ecology (1869, Ernst Haeckel) Ecosystem (1935, A.G. Tansley) se pehle aaya — dono foundational terms ke beech 66 saal ka gap hai.

2. Ecology vs Ecosystem — "H before T" Trick

- Term 'Ecology' sabse pehle use hua → Ernst Haeckel (1869).
- Term 'Ecosystem' sabse pehle use hua → A.G. Tansley (1935).

🧠 MNEMONIC — "Haeckel Higher (older), Tansley Trailing"

H se Haeckel, H alphabetically pehle aata hai aur 1869 mein bhi PEHLE aaya. T se Tansley, baad mein (1935) aaya, Ecosystem word diya. Alphabet order = time order — H-Ecology-1869 pehle, T-Ecosystem-1935 baad.

⚠️ EXAM TRAP

Haeckel aur Tansley ke naam/terms swap ho jaate hain. Fix: Ecology = Haeckel, Ecosystem = Tansley — dono alag log, alag decades (1869 vs 1935).

3. Components & Types

Biotic Components	Abiotic Components
Producers, Consumers (P/S/T), Decomposers (Bacteria & Fungi), Reducers	Organic, Inorganic, Climatic

- Terrestrial Ecosystem → Forest, Grassland, Desert (sabse zyada biomass).
- Aquatic Ecosystem → Marine (Permanent), River/Lake, Wetlands.

4. Food Chain & Trophic Levels — 10% Rule

Trophic Level	Examples
Producer	Green plants, algae, photosynthetic organisms
Primary Consumer	Herbivores — cow, goat, etc.
Secondary Consumer	Small carnivores — snake, cat, etc.
Tertiary Consumer	Khaata hai Primary + Secondary consumers ko
Quaternary Consumer	Top Carnivores — lion, tiger, vulture

🧠 MNEMONIC — Lindeman's 10% Rule

Sirf 10% energy hi ek trophic level se agle tak transfer hoti hai — 90% heat/respiration mein loss. "10 mein se 1" yaad rakho: Producer 100 units → Primary Consumer 10 units → Secondary 1 unit.

⚠️ EXAM TRAP

Humans ko sirf Primary Consumer mat samjho — Humans Primary AUR Secondary Consumer dono ki tarah act karte hain (veg + non-veg dono khaate hain).

✔ **QUICK RECALL**

Food Web = interconnected food chains ka NETWORK, ek single linear chain nahi.

Biomagnification → higher trophic levels mein toxic substances ka excessive concentration (DDT, BHC, 2-4D, Mercury).

5. Ecological Pyramid — "Energy Never Bends" Trick

- Concept diya gaya → Charles Elton (1933).

Type	Shape
Pyramid of Number	Upright (Aquatic) / Inverted (Terrestrial)
Pyramid of Biomass	Upright (Terrestrial) / Inverted (Aquatic)
Pyramid of Energy	Hamesha Upright

🧠 **MNEMONIC — "Energy Never Bends"**

Pyramid of Energy HAMESHA upright hota hai — ko exception nahi, kyunki energy higher levels par hamesha decrease hoti hai. Number aur Biomass "mood swing" karte hain (ecosystem type ke hisaab se invert ho sakte hain), Energy kabhi nahi.

⚠ **EXAM TRAP**

Number aur Biomass pyramids ke Upright/Inverted combos EXACTLY OPPOSITE hain: Number → Upright=Aquatic, Inverted=Terrestrial. Biomass → Upright=Terrestrial, Inverted=Aquatic. Energy ko in dono se confuse mat karo — wo hamesha fixed hai.

6. Niche & Imbalance

- Ecological Niche concept → Charles Elton (1927) & G.E. Hutchinson — organism ka structural aur functional role.
- Ecological Imbalance → species population mein sudden increase/decrease se adaptive capacity kamzor (Deforestation, Cyclone, Drought).

🧠 **MNEMONIC — Elton ka Double Role**

Charles Elton do jagah appear karta hai: 1927 mein Niche concept, 1933 mein Ecological Pyramid concept. Elton = "1927 Niche, phir 1933 Pyramid" — pehle role define kiya, phir uska pyramid banaya.

✔ **QUICK RECALL**

Highest Productivity — Terrestrial: Tropical Rainforest | Aquatic: Estuaries.

7. Biogeochemical Cycles — "GCOW vs PS" Trick

- Gaseous Cycles → Nitrogen, Carbon, Oxygen, Water Cycle.
- Sedimentary Cycles → Phosphorus Cycle, Sulphur Cycle.

🧠 **MNEMONIC — "GCOW roams free, PS stays grounded"**

Gaseous cycles (N-Carbon-Oxygen-Water = "GCOW") atmosphere mein FREELY move karte hain — gas ki tarah. Sedimentary (Phosphorus-Sulphur = "PS") rocks/soil mein GROUNDED rehte hain — inka gaseous phase nahi hota.

⚠ **EXAM TRAP**

Water Cycle ko GASEOUS classify kiya gaya hai (N, C, O ke saath) — isse Sedimentary samajhne ki galti

mat karo. Sirf Phosphorus aur Sulphur Sedimentary Cycles hain.

- Major Biomes: Tropical Evergreen Forest, Monsoon Forest, Savanna, Mediterranean, Temperate Grassland, Temperate Coniferous Forest, Tundra, Marine Pelagic.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Ecology term kisne diya — Haeckel ya Tansley?	Ecology = Ernst Haeckel (1869); Ecosystem = A.G. Tansley (1935)
Energy transfer rate kitna hai har trophic level par?	Sirf 10% (Lindeman's Law) — baaki 90% heat/respiration mein loss
Humans sirf Primary Consumer hain?	Nahi — Humans Primary AUR Secondary Consumer dono ki tarah act karte hain
Pyramid of Energy kabhi invert hota hai?	Nahi — Energy pyramid HAMESHA upright hota hai, koi exception nahi
Pyramid of Number: Upright kis ecosystem mein?	Aquatic mein Upright, Terrestrial mein Inverted (Biomass ka ulta)
Water Cycle Gaseous hai ya Sedimentary?	Gaseous (N, C, O ke saath) — sirf Phosphorus & Sulphur Sedimentary hain
Ecological Niche concept kisne diya?	Charles Elton (1927) & G.E. Hutchinson — Elton ne 1933 mein Pyramid bhi diya

Biodiversity

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Biodiversity = Earth par life ki variety — genes aur species se lekar poore ecosystems tak.
- Cover hota hai: levels (Alpha/Beta/Gamma), hotspots, in-situ & ex-situ conservation, IUCN Red List, Ramsar Convention + India ke 91 wetland sites.

✓ QUICK RECALL

Biodiversity equator se poles ki taraf DECREASE hoti hai — tropical regions (rainforests, coral reefs, islands) mein Earth ki highest biodiversity.

2. Levels of Biodiversity — ABG Trick

Type	Greek Letter	Description
Genetic Diversity	Alpha (α)	Organisms/species ke andar variation — Numerical
Species Diversity	Beta (β)	Species/communities ke beech diversity — Structural
Community Diversity	Gamma (γ)	Communities/landscapes ke beech inter-relationships

🧠 MNEMONIC — A-B-G = "Andar-Beech-Ghumte"

Alpha → ANDAR (species ke andar variation). Beta → BEECH (species ke BEECH diversity). Gamma → GHUMte-phirte landscapes ke beech relationships. A-B-G se A-Andar, B-Beech, G-Ghumta — order hi Greek alphabet order hai.

⚠️ EXAM TRAP

Gamma diversity ka concept WHITTAKER ne propose kiya tha — exam mein ise galat tarike se doosre ecologists ko attribute kar diya jaata hai. Whittaker = Gamma, yaad rakho.

✓ QUICK RECALL

Highest Biodiversity zones: Tropical Rainforests, Coral Reefs, Islands, Tropical Marine Zones. Largest Coral Reef → Great Barrier Reef (Australia) — "Rainforest of the Ocean".

3. Biodiversity Hotspots — Myers 1988

- Concept: jahan diversity zyada ho AUR us diversity ko threat bhi zyada ho.
- Diya gaya → Norman Myers (1988).

#	India's Hotspot
1	Western Ghats, Sri Lanka region
2	Eastern Himalaya region
3	Indo-Burma region
4	Sundaland, Nicobar Island group

🧠 MNEMONIC — "4 Hotspots = WEIS"

W(estern Ghats) - E(asterne Himalaya) - I(ndo-Burma) - S(undaland+Nicobar) — "WEIS" ek four-letter word ki tarah yaad karo, India ke 4 hotspots cover ho jaate hain.

4. Conservation Methods — Latin Root Trick

In-situ (On-site)	Ex-situ (Off-site)
National Parks, Wildlife Sanctuaries, Sacred Groves, Lakes	Biological Parks, Zoos, DNA Banks, Seed Banks

✓ QUICK RECALL

Latin roots directly hint karte hain: IN-situ = "andar" (natural habitat KE ANDAR) | EX-situ = "bahar" (natural habitat KE BAHAR).

5. IUCN Red List — 1978 vs 1988 Trick

Detail	Value
Full Form	International Union for Conservation of Nature
Established	1948
Headquarters	Gland, Switzerland
Red List — Plants	1978
Red List — Animals	1988

🧠 MNEMONIC — "Plants grow first (78), Animals follow (88)"

Plants ki Red List PEHLE aayi (1978) — jaise real life mein plants pehle exist karte hain, phir animals unhe khaate hain. Animals ki list ek dashak baad (1988). "78 se 88" — ek decade ka clean gap.

⚠️ EXAM TRAP

Plants (1978) aur Animals (1988) ke publication years ko SWAP mat karo — ye ek dashak (10 saal) apart hain, aur Plants hamesha pehle.

- Endangered Categories (decreasing severity): Extinct → Critically Endangered → Endangered → Vulnerable → Near Threatened.

🧠 MNEMONIC — "ECEVN" ladder

E-C-E-V-N: Extinct (sabse severe) → Critically Endangered → Endangered → Vulnerable → Near Threatened (sabse kam severe). Severity ladder neeche jaate jaate kam hoti hai.

Examples: Asiatic Cheetah, Pink-headed Duck, Indian Cheetah, Vultures, Nilgai, Golden Langur, Great Indian Bustard.

6. Ramsar Convention — "71 Sign, 82 Join" Trick

Detail	Value
Purpose	Marshy lands par wetlands ka conservation
Signed	2nd February, 1971 — Ramsar, Iran
Observance	World Wetland Day (2nd February)
India Joined	1982
Total Ramsar Sites in India	91 (as of June 2025)
Highest Sites — State	Tamil Nadu (20)
Largest Site	Sundarbans (West Bengal)
Smallest Site	Renuka (Himachal Pradesh)

 MNEMONIC — "11-year late entry"

Convention 1971 mein sign hui, lekin India 1982 mein hi join kar paya — 11-year gap. "71 + 11 = 82" — simple addition se yaad rakho.

 EXAM TRAP

Ramsar Convention SIGN hone ka saal (1971) aur INDIA ke JOIN karne ka saal (1982) alag hain — ye 11-year gap MCQ mein standalone question ki tarah frequently test hota hai.

 QUICK RECALL

Tamil Nadu akele 20 Ramsar sites rakhta hai — kisi bhi state mein sabse zyada, mostly bird sanctuaries, state ki wetland belt mein clustered.

7. Ramsar Sites — State-wise Reference (91 Sites)

State / UT	Ramsar Sites (Count)
Jammu & Kashmir	5 — Hokesar, Surinsar-Mansar, Wular, Haigam, Shallabugh
Ladakh	2 — Tso Kar Complex, Tsomoriri Lake
Himachal Pradesh	3 — Chanderal, Pong Dam, Renuka
Punjab	6 — Beas, Harike, Kanjli, Ropar, Keshopur-Miani, Nangal
Haryana	2 — Bhindawas, Sultanpur
Uttarakhand	1 — Asan Conservation Reserve
Uttar Pradesh	10 — incl. Sur Sarovar, Upper Ganga, Haiderpur
Sikkim	1 — Khecheopalri Wetland
Assam	1 — Deepor Beel
Manipur	1 — Loktak Lake
Tripura	1 — Rudrasagar Lake
Mizoram	1 — Pala Wetland
Rajasthan	4 — Keoladeo, Sambhar, Kichan, Menar
Gujarat	4 — Nalsarovar, Thol, Wadhvana, Khijadia
Madhya Pradesh	5 — Bhoj, Sakhya Sagar, Sirpur, Yashwant Sagar, New Reservoir
Maharashtra	3 — Lonar, Nandur Madhameshwar, Thane Creek
Goa	1 — Nanda Lake
Karnataka	4 — Ranganathittu, Ankasamudra, Asanasanini, Magadikere
Kerala	3 — Ashtamudi, Sasthamkotta, Vembanad-Kol
Jharkhand	1 — Udhwa Lake
Bihar	3 — Kabar Tal, Nagi, Nakti
West Bengal	2 — East Kolkata Wetlands, Sundarbans
Odisha	6 — Bhitarkanika, Chilika, Sakoshia, Tampura, Hirakud, Ansupa
Andhra Pradesh	1 — Kolleru Lake
Tamil Nadu	20 — largest count, mostly bird sanctuaries

 MNEMONIC — "UP-Punjab-TN" as top-3 anchor

State-wise पूरी list yaad karne ki jagah top-3 anchor karo: Tamil Nadu (20, No.1) → UP (10, No.2) → Punjab/Odisha (6, tied). Baaki states ke liye Largest=Sundarbans (WB), Smallest=Renuka (HP) do hi

extremes yaad rakho — ye MCQ mein most tested hain.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Gamma diversity ka concept kisne diya?	Whittaker (na ki koi aur ecologist — commonly misattributed)
In-situ ya Ex-situ — Zoo kaunsa hai?	Ex-situ (natural habitat ke BAHAR); National Park = In-situ
IUCN Red List Plants pehle aayi ya Animals?	Plants = 1978 (pehle); Animals = 1988 (ek dashak baad)
Ramsar Convention sign kab, India join kab?	Sign = 1971 (Ramsar, Iran); India Joined = 1982 (11-year gap)
Sabse zyada Ramsar sites kis state mein?	Tamil Nadu — 20 sites (sabse zyada)
Largest aur Smallest Ramsar Site India mein?	Largest = Sundarbans (WB); Smallest = Renuka (HP)
Biodiversity Hotspots concept kisne diya?	Norman Myers (1988) — India ke 4 hotspots: Western Ghats, Eastern Himalaya, Indo-Burma, Sundaland+Nicobar

Wildlife Conservation

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Cover hota hai: UNESCO Biosphere Reserves (18), Tiger Census, Wildlife Conservation Projects timeline, aur 58 Tiger Reserves ki state-wise list.

✓ QUICK RECALL

Project Tiger (1973) Jim Corbett National Park se launch hua — yahi park India ka FIRST National Park bhi tha (Hailey NP ke naam se, 1936).

2. Biosphere Reserves — "First ≠ Largest" Trick

- UNESCO ke Man and Biosphere Programme (1971) ke under shuru hua.
- Purpose: ecosystems aur unke genetic materials ka conservation.

Biosphere Reserve	State / Location
Nilgiri (FIRST, 1986)	Karnataka, Kerala
Great Rann of Kutch (LARGEST)	Gujarat
Dibru Saikhowa (SMALLEST)	Assam
Nanda Devi	Uttarakhand
Khangchendzonga	Sikkim
Sundarbans	West Bengal
Simlipal	Odisha
Gulf of Mannar	Tamil Nadu

🧠 MNEMONIC — "Nilgiri was FIRST, Kutch is BIGGEST"

Nilgiri (1986) = FIRST Biosphere Reserve — chota sa, history-maker. Great Rann of Kutch = LARGEST — bada sa, size-champion. Do alag superlatives, do alag reserves — "First" aur "Largest" ko kabhi mix mat karo.

⚠️ EXAM TRAP

Nilgiri (First) aur Great Rann of Kutch (Largest) ko swap karna sabse common mistake hai. Fix: N for Nilgiri = Number 1 (first); Kutch = King-size (largest).

✓ QUICK RECALL

Nagarhole National Park, Nilgiri Biosphere Reserve ka HI part hai — commonly tested linkage.

3. Important Facts — Park Linkages

- Asian Black Bear aur Snow Leopard → Nanda Devi aur Valley of Flowers National Parks.
- Estuarine Crocodile Conservation → Bhitarkanika Wildlife Sanctuary (Odisha).

🧠 MNEMONIC — "Hailey → Corbett → Ramganga" chain

India ka First National Park teen naam se guzra: Hailey National Park (1936, original naam) → Jim Corbett National Park (renamed) → 2021 mein Ramganga National Park bhi rename hua. Ek hi park, teen historical naam — chronological order mein yaad karo.

4. Tiger Census — Key Numbers

Detail	Value
Total Tiger Population	3,682
Total Tiger Reserves (2025)	58
India's Share of World Tigers	75%
Total Area Covered	75,769.83 km ²
Highest Tiger Population (order)	Madhya Pradesh > Karnataka > Uttarakhand
"Tiger State"	Madhya Pradesh

 MNEMONIC — "MP-Karna-Utta" order

Highest Tiger Population order yaad karo: Madhya Pradesh (Tiger State, No.1) → Karnataka (No.2) → Uttarakhand (No.3). "M-K-U" — teen letters, teen states, fixed order.

5. Conservation Projects Timeline — "Hangul Before Tiger" Trick

Project Name	Year
Hangul Project	1970
Project Lion	1972
Project Tiger	1973
Crocodile Conservation	1975
Manipur Thamin Project	1977
Project Rhino	1987
Project Elephant	1992
Project Red Panda	1994
Project Turtle	1999
Project Snow Leopard	2009
Project Great Indian Bustard	2013

 MNEMONIC — "70-72-73" ladder start

Hangul(70) → Lion(72) → Tiger(73) — teen saal ke andar teen projects, chronological ladder shuru hoti hai. Baaki projects (Croc 75, Rhino 87, Elephant 92, Red Panda 94, Turtle 99, Snow Leopard 09, GIB 13) isi ladder ko aage badhaate hain.

 EXAM TRAP

Hangul Project (1970) Project Tiger (1973) se 3 SAAL PEHLE aaya hai — lekin Tiger zyada famous hone ki wajah se log isko hi "first" samajh lete hain. Fix: Hangul is the actual first, launched 1970.

6. Tiger Reserves — "Biggest vs Smallest" Trick

State	Notable Tiger Reserves
Rajasthan	Ranthambore, Sariska, Mukundra Hills, Ramgarh Vishdhari
Madhya Pradesh	Kanha, Bandhavgarh, Panna, Pench, Satpura (9 reserves total)
Karnataka	Bandipur, Bhadra, Kali (Dandeli-Anshi), Nagarhole, BRT
Assam	Manas, Nameri, Kaziranga, Orang

State	Notable Tiger Reserves
Telangana	Kawal, Amrabad (BIGGEST Tiger Reserve)
Maharashtra	Melghat, Tadoba-Andhari (oldest in MH), Bor (SMALLEST Tiger Reserve)
Tamil Nadu	Kalakad-Mundanthurai, Annamalai (Indira Gandhi), Mudumalai
Chhattisgarh	Indravati, Udanti-Sitanadi, Achanakmar, Guru Ghasidas-Sanjay

 MNEMONIC — "Amrabad Big, Bor Small"

Amrabad (Telangana) = BIGGEST Tiger Reserve. Bor (Maharashtra) = SMALLEST Tiger Reserve. Dono alag states, opposite extremes — "A for Amrabad-Ache-bada (big), B for Bor-Bahut-chota (small)".

 EXAM TRAP

Amrabad (Biggest, Telangana) aur Bor (Smallest, Maharashtra) ko kis state mein hai ye mix karna common mistake hai. Ye ek bahut commonly asked pair hai — states swap mat karo.

 QUICK RECALL

Tadoba-Andhari Tiger Reserve, Maharashtra ka OLDEST Tiger Reserve hai.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Nilgiri First hai ya Largest Biosphere Reserve?	Nilgiri = FIRST (1986); Great Rann of Kutch = LARGEST (Gujarat)
India ka First National Park ka original naam?	Hailey National Park (1936) → Jim Corbett → 2021 mein Ramganga NP rename
Hangul Project ya Project Tiger — kaunsa pehle aaya?	Hangul Project (1970) — Project Tiger (1973) se 3 saal pehle
Sabse zyada Tiger Population wala state?	Madhya Pradesh (Tiger State) > Karnataka > Uttarakhand
Biggest aur Smallest Tiger Reserve kaunse hain?	Biggest = Amrabad (Telangana); Smallest = Bor (Maharashtra)
Estuarine Crocodile Conservation kahan hoti hai?	Bhitarkanika Wildlife Sanctuary (Odisha)
Total Tiger Reserves aur Population kitni hai (2025)?	58 Reserves, 3,682 tigers, India ka 75% world share

Climate Change

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Climate Change = weather conditions ke pattern mein historical changes — natural aur man-made dono causes.
 - Cover hota hai: causes & mitigation, Greenhouse Effect, Global Warming, Ozone depletion, India ke 7 National Missions, aur global climate conferences (Stockholm 1972 se Baku 2024 tak).
-  **MASTER CHAIN**
Stockholm (1972) → UNEP bani | Rio (1992) → UNFCCC ka janam | Kyoto (1997) → first binding targets | Paris (2015) → 2°C target | Glasgow (2021) → 1.5°C tak tighten.

2. Causes & Mitigation

Natural Causes	Man-made Causes
Solar flares, volcanic eruptions, atmospheric gas changes	Industrialization, deforestation, fossil fuel combustion, urbanization, agriculture

- Mitigation measures: Carbon Tax, Carbon Separation, Carbon Sink, Carbon Label, Earth Hour, Carbon Offset, Carbon Credit.

3. Greenhouse Effect vs Global Warming — "Natural vs Excess" Trick

 **MNEMONIC — "GHE ka Necessary, GW ka Excess"**
Greenhouse Effect = NATURAL, NECESSARY phenomenon — Earth ko life ke liye warm rakhta hai.
Global Warming = us effect ka HARMFUL EXCESS, rising GHG concentrations ki wajah se. "GHE good, GW too much" — synonyms mat samjho.

 **EXAM TRAP**
Greenhouse Effect aur Global Warming ko same cheez samajhna common mistake hai — GHE natural hai (zaroori), GW us natural process ka harmful over-extension hai.

CO₂ = major greenhouse gas. Methane (CH₄) source → Paddy fields, marshy areas. Doosri gases: Ozone, CFCs, Water Vapour, Nitrous Oxide, CO, SO₂.

 **QUICK RECALL**
Global Temperature ↑ → Photosynthesis rate ↓ — ye ek direct inverse relationship hai.

4. Ozone Layer — "85 Discover, 87 Protocol" Trick

Detail	Value
Location	Stratosphere — Earth surface se 50 km tak
Max Concentration	15–35 km altitude
Function	UV rays se living beings ko protect karta hai
Measured in	Dobson Units
Main Cause of Depletion	Chlorofluorocarbons (CFCs)
Ozone Hole Discovered	Antarctica — 1985
Ozone Layer Protocol	Montreal Protocol — 1987
Ozone Protection Day	16th September

 MNEMONIC — "85 dekha, 87 bachaya"

1985 mein Antarctica mein ozone hole DEKHA gaya (discover). 1987 mein Montreal Protocol se use BACHANE ki koshish shuru hui — 2 saal ka gap. "Dekha pehle, bachaya baad mein."

 EXAM TRAP

Ozone hole DISCOVERY (1985, Antarctica) aur Montreal PROTOCOL (1987, protection) ke years ko confuse mat karo — 2-year gap hai, discovery hamesha pehle.

5. National Missions — "2010 Cluster" Trick

Mission	Year
National Mission on Enhanced Energy Efficiency	2008
National Solar Mission	2010
National Mission on Strategic Knowledge for Climate Change	2010
National Sustainable Agriculture Mission	2010
National Mission on Sustainable Habitat	2010
National Water Mission	2011
National Green India Mission	2014
NMSHE (Himalayan Ecosystem)	2014

 MNEMONIC — "2010 Char" (2010's Four)

8 missions mein se 4 SAME SAAL (2010) mein launch hui: Solar, Strategic Knowledge, Sustainable Agriculture, Sustainable Habitat — sab "S" se start hone wale naam (Solar, Strategic, Sustainable×2). "2010 ka S-quad" yaad rakho.

6. Climate Conferences — Full Chronology

Conference	Year	Key Point
Stockholm	1972	UNEP ki establishment
Vienna	1985	Ozone Layer Protection
Montreal Protocol	1987	Ozone Protection Day (16 Sept)
Rio (Earth Summit-I)	1992	Agenda-21; UNFCCC ka janam
Kyoto Protocol	1997	GHG 5% kam (1990 levels se)
Johannesburg (Rio+10)	2002	Sustainable dev.; World Solidarity Fund
Montreal (2nd)	2005	Developed countries GHG kam
Bali	2007	2050 tak GHG 2000 levels par
Copenhagen	2009	Low-carbon economy concept
Cancun	2010	Green Climate Fund establish
Durban (CoP-17)	2011	Green Climate Fund strengthen
Doha (CoP-18)	2012	CDM ke under pollution kam
Warsaw (CoP-19)	2013	Green Climate Fund agreement
Lima (CoP-20)	2014	2070 tak GHG kam karne ka plan
Paris (CoP-21)	2015	2100 tak warming <2°C

Conference	Year	Key Point
Marrakesh (CoP-22)	2016	Clean water, water management
Bonn (CoP-23)	2017	Financial assistance, mitigation
Katowice (CoP-24)	2018	2020 mein Paris Agreement implement
Madrid (CoP-25)	2019	GHG emissions par deliberation
Glasgow (CoP-26)	2021	Warming 1.5°C se zyada na ho
Sharm El-Sheikh (CoP-27)	2022	GHG mein reduction
Abu Dhabi (CoP-28)	2023	\$100 billion annual funding
Baku (CoP-29)	2024	Carbon market agreement, methane kam



MNEMONIC — "Do Montreal" — city repeat trick

Montreal naam se DO alag events hain: 1987 = Montreal PROTOCOL (ozone protection) aur 2005 = Montreal CONFERENCE (developed countries GHG reduction). Same city, alag saal, alag purpose — merge mat karo.



EXAM TRAP

Montreal (1987, ozone) aur Montreal (2005, GHG) — dono alag conferences hain jo sirf city-naam share karte hain. Exam mein inhe ek hi event samajh lena common trap hai.



QUICK RECALL

CoP numbering Durban (CoP-17, 2011) se clearly shuru hoti hai zyadatar references mein — usse pehle ki conferences (Rio, Kyoto, Bali, Copenhagen, Cancun) ko city-naam se hi refer kiya jaata hai, CoP number se nahi.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Greenhouse Effect aur Global Warming same hain?	Nahi — GHE natural/necessary hai, GW us effect ka harmful excess hai
Ozone hole kab discover hua, protocol kab bana?	Discover = 1985 (Antarctica); Montreal Protocol = 1987 (2-year gap)
2010 mein kitni National Missions launch hui?	4 — Solar, Strategic Knowledge, Sustainable Agriculture, Sustainable Habitat
Do Montreal Conferences mein kya difference hai?	1987 Montreal = ozone protocol; 2005 Montreal = developed countries GHG reduction
UNFCCC ka janam kaunse conference mein hua?	Rio Earth Summit-I (1992), saath hi Agenda-21 approve hua
Paris Agreement ka target kya tha?	2100 tak global warming 2°C se neeche (CoP-21, 2015)
CoP numbering kaunse conference se shuru hoti hai?	Durban (CoP-17, 2011) se — usse pehle city-naam se refer hota hai

Environmental Pollution

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Cover hota hai: Air, Water, Soil, Noise, Radioactive, Solid Waste pollution — causes, impacts, control measures.
- Special focus: disease-pollutant matching (Minamata–Mercury, Itai-Itai–Cadmium, etc.) — bahut common MCQ pattern.

✓ QUICK RECALL

Disease-Cause matching: Minamata → Mercury | Itai-Itai → Cadmium | Black Foot → Arsenic | Blue-baby Syndrome → Nitrates.

2. Primary vs Secondary Pollutants — "Direct vs Reaction" Trick

Primary Pollutants	Secondary Pollutants
DDT, plastic, CO, CO ₂ , NO _x , SO ₂ , particulate matter	O ₃ , NH ₃ , HNO ₃ , H ₂ SO ₄ , SO ₃ (Acid Rain jaisi reactions se banti hain)

🧠 MNEMONIC — "Primary = Direct Emit, Secondary = Sky-Cooked"

Primary pollutants SEEDHE emit hote hain (chimney se CO seedha nikalta hai). Secondary pollutants atmosphere mein PRIMARY pollutants ke reactions se "cook" hote hain — jaise Ozone aur Acid Rain. "Direct nikla = Primary, Sky mein bana = Secondary."

⚠️ EXAM TRAP

Ozone (O₃) ya Acid Rain components (H₂SO₄, HNO₃) ko Primary Pollutant samajhna common mistake hai — ye SECONDARY hain, atmosphere mein reaction se bante hain.

3. Air Pollution

- Causes: Fossil fuels ka incomplete combustion, industrial waste, vehicles.
- Pollutants: Gaseous (CO, CO₂, SO₂, NO_x) | Particulate (aerosols, smoke, soot, smog).
- Impacts: Asthma, respiratory diseases, brain/kidney disorders, lung cancer.

✓ QUICK RECALL

Government efforts: Air (Prevention and Control of Pollution) Act, 1981 aur National Air Quality Monitoring Programme.

4. Water Pollution — Disease-Agent Matching Table

Disease	Cause / Agent
Jaundice	Virus
Minamata Disease	Mercury
Blue-baby Syndrome	Nitrates
Cholera, Typhoid, Dysentery	Bacterial
Itai-Itai Disease	Cadmium
Black Foot Disease	Arsenic

MNEMONIC — "MICA" Metal-Disease Trick

M-inamata → M-ercury (dono M se shuru). I-tai-Itai → C-adium (Itai-Itai mein do baar "I" hai, lekin agent Cadmium hai — is exception ko yaad rakho). B-lack Foot → A-rsenic (B aur A adjacent letters). "MICA": Mercury-Itai(Cadmium)-Cadmium-Arsenic — ek group mein confuse hone wale 4 pairs.

EXAM TRAP

Minamata (Mercury) aur Itai-Itai (Cadmium) ko SWAP karna sabse common trap hai — dono heavy-metal diseases hain, alag-alag metals. Fix: miNAMAta = merCURY nahi lagta dikhne mein, isliye ratta maaro: "Mina-Mercury", "Itai-Cadmium".

QUICK RECALL

BOD (Biochemical Oxygen Demand) = decomposition ke liye required oxygen — higher BOD = zyada organic pollution.

Government efforts: Ganga Action Plan (purana) vs Namami Gange Programme (current, comprehensive) — dono ka target same (Ganga conservation) lekin alag eras ki alag schemes.

5. Soil Pollution

- Causes: Chemicals, soil erosion, fertilisers ka overuse, industrial/urban waste.
- Measures: Chemical fertilisers par control, bioremediation, nuclear explosions par ban.
- Impacts: Soil properties ka loss, fertility decline, diseases ka spread.

6. Noise Pollution — "Silent Zone = Lowest" Trick

- Noise Pollution = 65 dB se zyada sound intensity.

Area Type	Day (dB)	Night (dB)
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silent zone	50	40

MNEMONIC — "75-65-55-50" descending ladder

Day-time limits ek clean descending pattern follow karte hain: Industrial(75) → Commercial(65) → Residential(55) → Silent(50). Har category 10 dB kam hoti hai, except Silent zone jo sirf 5 dB kam hota hai. Night limits bhi same order mein 5 dB kam: 70-55-45-40.

EXAM TRAP

Silent Zone ke permissible limits SABSE KAM hain (50 dB day / 40 dB night) — Residential area se bhi kam. Exam mein ye frequently swap ya Residential se confuse kar diya jaata hai.

7. Radioactive Pollution — "Bone vs Blood" Trick

Natural Sources	Man-made Sources
Solar, cosmic, terrestrial (rock, soil)	Nuclear testing, weapons, reactors, uranium mining

MNEMONIC — "Strontium loves Bones, Iodine loves Blood"

Strontium-90 → BONE cancer (Strontium chemically Calcium jaisa hai, isliye bones mein jama hota hai). Iodine-131 → RBC (Red Blood Cell) damage. "S for Strontium-Skeleton, I for Iodine-blood".

EXAM TRAP

Strontium-90 (Bone cancer) aur Iodine-131 (RBC damage) ke impacts ko swap mat karo — ye ek common MCQ pair hai.

Impacts: genetic mutations, cancer, tumor, infertility — DNA/RNA/protein structure affect hoti hai.

8. Solid Waste Pollution

- Sources: Urban/Rural/Domestic/Industrial Waste, Nuclear Plant Waste, Organic Waste.
- Disposal Methods: Landfill, composting, vermiculture, pyrolysis.
- E-Waste → TV, fridge, computer, mobile se waste. Incinerators → garbage jalaane ka modern process.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Ozone (O ₃) Primary hai ya Secondary Pollutant?	Secondary — atmosphere mein primary pollutants ke reactions se banta hai
Minamata Disease kis metal se hoti hai?	Mercury (Itai-Itai se confuse mat karo — wo Cadmium hai)
Itai-Itai Disease kis metal se hoti hai?	Cadmium (Minamata se confuse mat karo — wo Mercury hai)
Black Foot Disease kis element se hoti hai?	Arsenic
Silent Zone ka noise limit Residential se zyada hai?	Nahi — Silent Zone SABSE KAM hai (50 dB day / 40 dB night), Residential se bhi kam
Strontium-90 aur Iodine-131 ke impacts kya hain?	Strontium-90 = Bone cancer; Iodine-131 = RBC damage
Ganga Action Plan aur Namami Gange same hain?	Nahi — dono alag-alag eras ki alag schemes hain, target same (Ganga conservation)

Sustainable Development

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Sustainable Development = future generations ko dhyaan mein rakhte hue natural resources ka minimum exploitation — social, economic, environmental development saath mein.
- 17 SDGs 2015 mein determine hui, 1 Jan 2016 se effective hui, target completion 2030.

MASTER CHAIN

Determined (2015) → Effective (1 Jan 2016) → Target (2030) — determination aur implementation ke beech 1 saal ka gap, target tak 15 saal ka runway.

2. Core Concept — "Not Just Environment" Trick

- Natural resources ka minimum exploitation, future generations ko dhyaan mein rakhte hue.
- Shamil hai: Social development, Economic development, Environmental development.

EXAM TRAP

Sustainable Development sirf ENVIRONMENTAL protection ke baare mein NAHI hai — ye equally SOCIAL aur ECONOMIC development ko bhi cover karta hai. One-liner questions mein ye common conceptual trap hai.

MNEMONIC — "SEE" Triangle

Sustainable Development ke teen pillars: S-ocial, E-conomic, E-nvironmental — "SEE" (dekho) teeno equally important hain, sirf environment nahi.

3. SDGs 1-9 — "Basic Needs" Group

#	Goal
1	No Poverty
2	Zero Hunger
3	Good Health and Wellbeing
4	Quality Education
5	Gender Equality
6	Clean Water and Sanitation for All
7	Affordable and Clean Energy
8	Decent Work and Economic Growth
9	Industry, Innovation and Infrastructure

MNEMONIC — 3-3-3 Split (1-9)

Number 1-9 ko teen-teen ke groups mein todo: (1,2,3) Poverty-Hunger-Health = SURVIVAL basics. (4,5,6) Education-Gender-Water = SOCIAL equity. (7,8,9) Energy-Work-Industry = ECONOMIC engine. Teen groups, teen-teen goals — survival → social → economic, ek logical progression.

4. SDGs 10-17 — "Systems & Planet" Group

#	Goal
10	Reduce Inequalities
11	Sustainable Cities and Balanced Communities
12	Responsible Consumption and Production
13	Climate Action
14	Life Below Water
15	Life on Land
16	Peace, Justice and Strong Institutions
17	Partnerships for the Goals

MNEMONIC — "14-15 Ecosystem Pair"

SDG 14 (Life BELOW Water) aur SDG 15 (Life ON Land) ek "ecosystem pair" banate hain — dono directly biodiversity conservation se deal karte hain, ek AQUATIC (14, neeche/below), ek TERRESTRIAL (15, upar/on land). Number order hi domain order hai: pehle paani, phir zameen.

EXAM TRAP

SDG 14 aur 15 ke naam ("Life Below Water" vs "Life on Land") aur unke numbers ko swap karna common mistake hai — 14 = Water (chota number, neeche), 15 = Land (agla number, upar).

QUICK RECALL

SDG 17 (Partnerships for the Goals) sabse AAKHRI goal hai — ye sabhi 16 goals ko achieve karne ke liye "global partnership/collaboration" ka goal hai, isliye list ke end mein aata hai.



TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
SDGs kab determine hui aur kab effective hui?	Determined = 2015; Effective = 1 Jan 2016 (1-year gap); Target = 2030
Sustainable Development sirf environment ke baare mein hai?	Nahi — Social + Economic + Environmental teeno equally shamil hain
SDG 14 aur 15 mein kya farak hai?	SDG 14 = Life Below Water (aquatic); SDG 15 = Life on Land (terrestrial)
Total SDGs kitni hain?	17, target completion year 2030
SDG 9 kya cover karta hai?	Industry, Innovation and Infrastructure ka development
Last SDG (17) kya hai?	Partnerships for the Goals — sabhi 16 goals achieve karne ke liye global collaboration

Disaster Management

MEMORY-TRICK SHEET

1. Chapter Snapshot

- Cover hota hai: Earthquake, Tsunami, Landslide, Flood, Drought, Cyclone — India ke institutional disaster management framework ke saath.
- Saath mein: Cloudburst, Bhopal Gas Tragedy, Chernobyl Disaster bhi.

✓ QUICK RECALL

Disaster Management Act, 2005 ne three-tier structure establish kiya: National, State, aur District Disaster Management Authorities.

2. Earthquake Zones — "Delhi is NOT Safe" Trick

- Earthquake = Earth ke crust ki tectonic vibrations. Causes: plate tectonics, Earth ki rotation, volcanic activity, aquatic load.

Zone	Risk Level	Key States/Regions
Zone-2	Minimum (Intensity ≤ 6)	N & E Rajasthan, Tamil Nadu, N Andhra Pradesh, Chhattisgarh
Zone-3	Moderate (Intensity VII)	Kerala, Goa, Gujarat, UP, West Bengal, Odisha
Zone-4	High (Intensity VIII)	Jammu & Kashmir, Himachal Pradesh, Delhi, Sikkim
Zone-5	Very High (Intensity IX+)	North-East India, Uttarakhand, Western HP, Andaman Islands

🧠 MNEMONIC — "2-3-4-5 Rising Risk Ladder"

Zone number JITNA BADA, risk UTNA ZYADA — simple direct relationship: Zone-2 (kam) → Zone-3 → Zone-4 → Zone-5 (sabse zyada). Seismic zone number khud hi risk level ka indicator hai — koi reverse ya alag scale nahi.

⚠️ EXAM TRAP

Delhi ek PLAINS capital city hai, isliye log ise low-risk zone mein maan lete hain — lekin Delhi Zone-4 (HIGH impact) mein aata hai. "Capital $\neq$ Safe" — ye ek bahut commonly tested exception hai.

3. Tsunami & Landslide

- Tsunami ek JAPANESE word hai. Causes: Earthquake, volcanic eruption, submarine landslide.
- Management: Sea walls, mangrove plantation, Indian Tsunami Early Warning System (2007).
- Landslide: gravity ki wajah se rocks ka downward falling — Himalayan regions mein common.
- Landslide Management: Afforestation, mining par ban, contour farming.

4. Flood & Drought — "77 Before 90" Trick

- Flood: Rivers/lakes mein water ka sudden rise — heavy rainfall, siltation, faulty irrigation se.
- Drought (Atmospheric Disaster): Lambe samay tak low rainfall, groundwater decrease, water scarcity.

Programme	Year
Desert Development Programme	1977–78
National Watershed Development Project	1990–91

 MNEMONIC — "Desert First, Watershed Later"

Desert Development Programme (1977-78) National Watershed Development Project (1990-91) se EK DASHAK SE ZYADA PEHLE aaya. "D for Desert, D for Decade-pehle" — Desert programme hamesha pehle yaad karo.

 EXAM TRAP

Desert Development Programme (1977-78) aur National Watershed Development Project (1990-91) ke order ko SWAP mat karo — 13-year gap hai, Desert programme pehle aaya.

5. Cyclone & Disaster Management

- Cyclone: low pressure centre — Temperate aur Tropical dono regions mein, Tropical cyclones zyada destructive.
- Protection: Coastal areas mein mangrove vegetation promote karna.

 QUICK RECALL

Disaster Management Act, 2005 ke under National, State aur District Disaster Management Authorities ban'e.

6. Other Disasters — "1984 vs 1986" Trick

Disaster	Details
Cloudburst	Hilly areas mein sudden heavy rainfall — flood problems cause karta hai
Bhopal Gas Tragedy	Bhopal, 3rd December 1984 — Methyl isocyanate gas leak
Chernobyl Disaster	Ukraine, 26th April 1986 — Industrial nuclear incident

 MNEMONIC — "84 Gas, 86 Nuclear"

Bhopal Gas Tragedy (India, 1984) aur Chernobyl Disaster (Ukraine, 1986) sirf 2 SAAL KE ANTAR mein hue — dono world history ke landmark man-made industrial disasters. "84 se 86" — 2-year gap, Bhopal pehle.

 EXAM TRAP

Bhopal (1984) aur Chernobyl (1986) ke countries aur years ko mix mat karo — Bhopal = India/Gas leak, Chernobyl = Ukraine/Nuclear incident, sirf 2 saal apart.

TOP TRAPS — FINAL RECAP

Common Confusion / Trap	Correct Fact
Delhi kaunse Earthquake Zone mein aata hai?	Zone-4 (High impact) — plains capital hone ke bawajood, low-risk NAHI
Zone number aur risk level ka relationship kya hai?	Direct — Zone number jitna bada, risk utna zyada (Zone-5 sabse risky)
Desert Development Programme ya National Watershed — kaunsa pehle?	Desert Development Programme (1977-78) — Watershed (1990-91) se 13 saal pehle
Bhopal Gas Tragedy kab hui, Chernobyl kab hua?	Bhopal = 1984 (India, gas leak); Chernobyl = 1986 (Ukraine, nuclear) — 2-year gap
Indian Tsunami Early Warning System kab establish hua?	2007
Disaster Management Act kab bana aur kya structure diya?	2005 — National, State, District Disaster Management Authorities
Tsunami kis language ka word hai?	Japanese